

University of Toronto
Faculty of Applied Science and Engineering
APS112
Project Requirements (PR)

Project #	013	Date	March 3rd, 2026
-----------	-----	------	-----------------

Project Title	Buzzer beaters: A portable buzzer system
Client	Alexandre Klaus
Tutorial Section	TUT0116
Teaching Assistant	Ash Mahmood
Communication Instructor	Ebru Bag
Engineering Manager	Glenn Hibbard
Prepared By (Names and Student #s of Team Members)	Omotayo Faseemo Merrick (MingXuan) Wang Henry (Yeonjun) Kim Abdulrahman Abuloghod Kamilia Desroches Parchment Shannon Ho

State your word count (excludes Cover Page, Executive Summary, References and Appendices).

☐

Have you verified that your team has included all required sections described in the Course Assignment Reference Handbook? [Y or N]

☐

Have you submitted your completed and signed Attribution Table PDF as a separate submission to Quercus? [Y or N]

☐

Have you used Generative AI (or Large Language Model) in the writing of this report? [Y or N] If yes, you must follow the GenAI usage and documentation course policy.

If any required components or sections are missing (including cover page or appendices), your assignment is considered incomplete, and standard course late penalties will accrue until the submission is complete.

Executive Summary

The Skule™ Kup is an annual University of Toronto engineering competition that involves nine engineering disciplines and multiple rapid-response games. In this competition, multiple participants attempt to respond simultaneously, so a reliable method is needed to determine the first responder. Currently there is no objective response detection system, so event organizers rely on subjective judgement. This leads to operational inefficiency and unfairness among participants, which reveals a clear gap between the demands of competition and existing practices.

The client, Alexandre Klaus, the current Skule™ Kup Director and the second-year Engineering Science student, identified the need for a reliable method to detect and identify the first responder while ensuring portability, safety, and ease of deployment across campus venues. Thus, the need is defined as a multi-user response detection system capable of accurately prioritizing simultaneous inputs under competitive conditions.

The scope of this project is narrowing to Skule™ Kup's events, primarily held in The Pit and the EngSoc Arena. The system must be capable of supporting 2 to 8 participants and operate in an indoor service environment where moderate to high ambient noise levels, frequent setups occur, placement at approximately ~1 meter height is required, potential drop hazards exist during operation, and limited storage space is available.

Key interest holders include the Skule™ Kup Director, incoming Director, participants and organizers of the competitions, the University of Toronto Facilities & Services Department, and the Engineering Society (EngSoc). They influence a wide range of areas, from regulatory approvals and funding management to operational usability and competition fairness. Through client meetings, operational concerns were converted into measurable engineering requirements.

The primary function of this system is to detect and display the first response from multiple users simultaneously. Secondary functions include capturing and transmitting user input, providing direct output to the host, enabling safe handling, and supporting repetitive setups. These functions are supported by ranked objectives including lightweight design (≤ 0.5 kg per buzzer), compact packaged dimensions ($\leq 25(L) \times 30(W) \times 23(H)$ cm), impact resistance (1 m drop), button lifecycle durability ($\geq 100,000$ cycles), MYFab manufacturability ($\geq 80\%$ in-house fabrication), cost limitations ($\leq \$250$ CAD), and establishment of the Skule™ visual identity.

The design must also satisfy these constraints, including a supporting interaction with a minimum of 2 users, a minimum two-hour of continuous runtime, compliance with the Ontario Electrical Safety Code (OESC) – safe electrical insulation with zero exposed components, voltage regulation within $\pm 5\%$, and operation below 80% of operating current. Storage dimensions must not exceed the allocated Skule™ locker limits ($\leq 50(L) \times 50(W) \times 95(H)$ cm).

This report synthesizes the identified gaps, service environment, interest holder requirements, functions, objectives, and constraints into a systematic problem definition framework. By clearly connecting with customer requirements to quantified metric criteria, it establishes a foundation for evaluating conceptual design alternatives that balance operational performance, durability, safety, and portability, and for selecting the optimal solution for Skule™ Kup competitions.

1.0. Introduction

The client for this project is Alexandre Klaus, a second-year Engineering Science student at the University of Toronto and the current Skule™ Kup director [1]. Skule™ Kup is a year-long competition between engineering disciplines, designed to foster friendly-rivalry through various games. This includes Discipline Feud, where teams must guess the most common answers to survey questions. Players compete by alerting the game host when they wish to answer, with priority given to those with the fastest timing. The client aims to improve the detection of distant information in Discipline Feud competitions, while maintaining the system's safety and portability.



Figure 1. The pit in the basement, Myhal Centre for Engineering Innovation and Entrepreneurship - where the Skule™ Kup event will be held [2].

This report will examine existing buzzer system designs, define the environment in which the design will be used, analyze the groups of people impacted by this project, identify goals and restrictions, and finally state the design's main functions.

2.0. Problem Statement

Alexandre Klaus has identified operational challenges in managing rapid-response competitions, where multiple contestants often respond simultaneously. The gap is the lack of a reliable method to identify first responders in Skule™ Kup competitions, predominantly Discipline Feud. This can quickly lead to disputes and interruptions as organizers must rely on subjective judgment to determine the response order. Consequently, this reduces both efficiency and perceived fairness of the competition. There is a need for a method of first responder detection within Discipline Feud sessions, capable of processing a variable number of contestants in real time.

The scope of the project is mainly confined to Skule™ Kup's Discipline Feud events, which are most commonly held in The Pit—a students' common space located in the basement of Sandford Fleming. This scope was defined through consultation with Alexandre Klaus, who emphasized this event type and setting as the intended context for the solution's use. The intended users include event participants, primarily students and alumni, as well as the Skule™ Kup director responsible for facilitating the competition. The design will function under typical event operations, which include frequent handling, temporary setup, and transportation between storage and venues.

3.0. Service Environment

This upcoming section elaborates on the conditions in which the system will operate within the renovated Pit and Engineering Society (EngSoc) Arena. Physical and virtual attributes come before living things with The Pit specifications before EngSoc Arena's. All tables are ordered from most to least important.

3.1. *Physical*

The Pit is a student lounge in the Sanford Fleming Building at the centre of student life that's currently under renovations so specifications were unavailable [3]. Below is Table 1.

Table 1. Physical environment of the Pit

Component	Description
Atrium/seating (tables, benches, and decks) [3]	Provides a gathering space for the game
Accessibility upgrades	Ramps will be utilized to promote inclusion [3] [4].
Lighting	A/V lighting system will keep the area well lit [4].
Temperature	Temperatures are expected to be $\sim 21^{\circ}\text{C}$ (Appendix A).



Figure 2. 3D rendering of the renovated Pit atrium [5].

The EngSoc Arena is a student design space in the Myhal Centre basement being used as a temporary space for Skule Kup [2]. Below is Table 2.

Table 2. Physical environment of the EngSoc Arena

Component	Description
Atrium / seating (tables, benches, and decks)	~9 tables of 0.85m and 1m heights with ~20 chairs (Appendix A)
Sound level	The sound during a site visit was 46 dB with a peak frequency of 23 Hz as measured by the Decibel Meter app (Appendix A). Ambient volume is likely to increase during competition.
Lighting	The area's lighting is controlled by a six zone touchscreen Leviton unit (Appendix A).
Temperature	Temperatures are expected to be ~21°C (Appendix A).

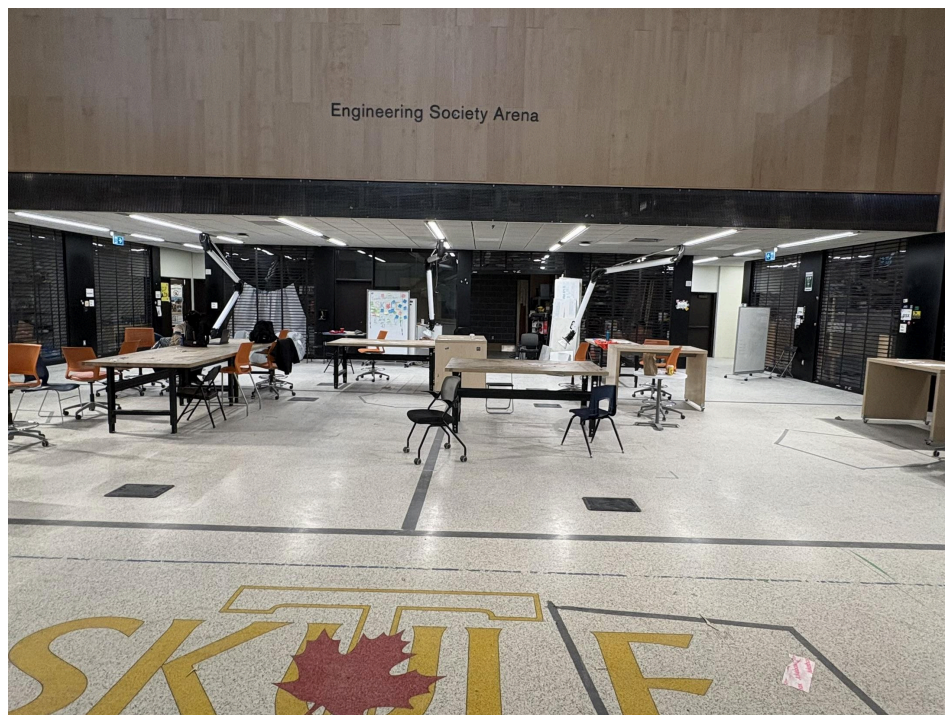


Figure 3. Current layout of the EngSoc Arena.

3.2. *Living Things*

People who aren't users within the same space as the game may impact the design of our solution. The Pit renovations prevented the collection of passersby data. Below is Table 3.

Table 3. Living things within The Pit and EngSoc Arena

Component	Description
Small birds	
Students	At the EngSoc Arena during the third site visit were seven/seven U of T engineering students, four girls and three guys (Appendix A).
Staff	

3.3. *Virtual*

Discipline Feud is to be held indoors either in the Myhal Centre or the Sandford Fleming building so the appropriate digital components were identified (Appendix A). Below is Table 4.

Table 4. Virtual environment of The Pit and EngSoc Arena

Component	Description
Electrical availability	The client confirmed outlets are available (Appendix A).
Wifi connection	Wifi is provided throughout campus through the UofT network [6].
Cellular/Data connection	Telus, Rogers, and Bell provide a 5G network across campus [6], [7], [8].

4.0. Interest Holders

Our design involves multiple interest holders who either influence its implementation or are directly impacted by its use during Skule™ Kup events, as shown in Table 4. All interest holders are ranked from most to least important with justification shown in Appendix C.

Table 4. Identified Interest holders & Justification

Interest Holder	Justification
University of Toronto Facilities & Services	Highest influence. UofT Facilities& Services has regulatory authority to approve or deny the use of temporary equipment in campus spaces, which determines whether the system can be deployed [9].
University of Toronto Engineering Society (EngSoc)	High Influence. EngSoc provides funding for Skule™ Kup and oversees event execution, so its approval affects whether the system is supported and used (Appendix B).
Future Skule™ Kup director	High influence. The future director controls how event equipment is managed and adopted in subsequent years (Appendix A) (Appendix B).
Discipline Feud game host	High impact. The host directly uses the system to identify and announce responses during the game and is affected by its performance, but does not control funding or approval (Appendix A).
Discipline Feud participants	Moderate impact. Competition outcomes depend on the system's response recognition, and participants are affected by its performance during the game, but they have no control over funding or implementation decisions [1].

5.0. Detailed Requirements

This section will include the functions, both primary and secondary, as well as the objectives and constraints of our design. These are aimed to inform the reader of the core problem the design seeks to solve, goals the system may accomplish, and set boundaries the design must not exceed.

5.1. Functions

As shown in Figure 4, the design functions are categorized as primary and secondary. Secondary functions enable and support the primary capabilities, which establish the design's intended purpose. Consult Appendix D for further justifications.

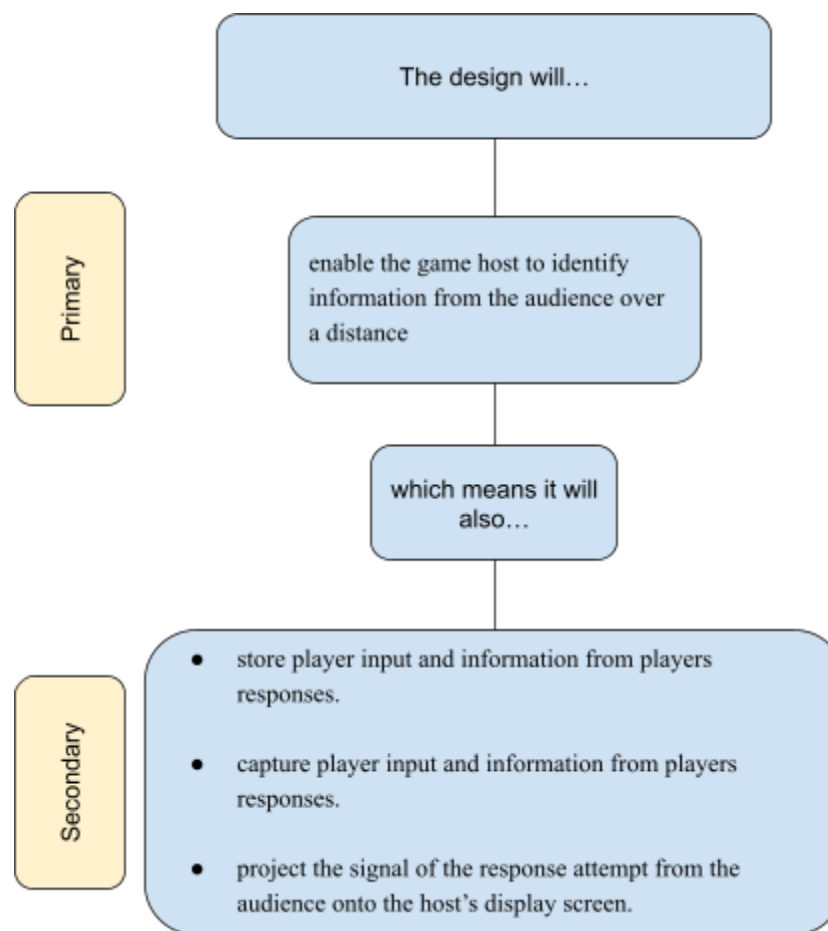


Figure 4. Primary and secondary functions of the design (53 words).

5.2. Objectives

The goals that the design should accomplish are outlined in the following objectives table. Appendix E contains more developed justifications for said objectives.

Table 5. Objectives ranked from most to least important.

	Descriptor	Metric	Goal	Justification
O.1.	Should be Lightweight	Weight per buzzer (kg)	≤ 0.5 kg [10][11]	Handheld devices such as gaming controllers and buzzers typically weigh 0.2 to 0.5 kg [10][11], preventing user fatigue. Goals are averaged from comparable commercial devices.
		Weight of the buzzer system (kg)	≤ 1.6 kg [10][11]	
O.2.	Compact in size	Maximum packaged dimensions: L x W x H (cm)	$\leq 25 \times 30 \times 23$ cm [12]	Air Canada carry-on limits ($\cong 25 \times 15 \times 43$ cm)[12] ensure the system fits in a locker and is portable by one person, as requested by the client (App.A)
O.3.	Impact resistant	Drop resistance (meters) without damage	1m drop [13]	IEC 60068-2-31 sets a 1 m drop test standard for portable electronics [13], which also corresponds to the approximate table height in the service environment.
O.4.	Should be durable	Button lifecycle (cycles)	$\geq 100,000$ cycles [14]	Commercial tactile switches are commonly rated for $\leq 100,000$ actuations before mechanical failure, as specified by manufacturers such as Omron and Panasonic.[14]
O.5.	Structurally resilient	Material with high impact strength (kJ/m^2)	Materials such as ABS or Polycarbonate with impact strength $\approx 60 - 80$ kJ/m^2 . [15]	ABS and polycarbonate are used in electronics housings for their high impact resistance, with MatWeb reporting impact strengths of 60–80 kJ/m^2 for these materials.[15]
O.6.	Manufacturable using MYFab materials	% of structural components fabricated in MYFab	$\geq 80\%$	Client specified the design should predominantly use MYFab materials (App.A). An 80% fabrication goal aligns with recommendations to maximize in-house fabrication.

O.7.	Remains within budget constraints	System cost (CAD)	$\leq$ \$250 (Appendix A)	Comparable 4–8 player commercial buzzer systems retail for \$200–\$400. [18]
O.8.	Visually consistent with Skule™ branding	Exterior colour compliance (%)	$\geq$ 80% visible surfaces in Navy blue and Gold [16]	Research shows that dominant colour coverage above 70% improves brand recognition [16], and navy and gold are the established colours of Skule™ identity.

5.3. Constraints

The solution must adhere to the following constraints, derived through consideration of the client's needs, project scope and governing Ontario regulations. Any designs failing to meet these requirements will be excluded from the solution space. Appendix F contains more detailed justifications for said constraints.

Table 6. Constraints that the design should adhere to.

	Descriptor	Metric	Limit	Justification
C.1.	Must support simultaneous use by a minimum of two users.	Number of users that can interact simultaneously.	≥ 2 users	Skule™ Kup competitions require multiple participants per session (Appendix G).
C.2.	Must operate for at least two hours before power depletion.	Continuous runtime, from full capacity until power depletion (hrs).	≥ 2 hrs	Discipline Feud sessions run for approximately two hours (App.A). Failure to meet this constraint would make the system unsuitable for competition use, as recharging or replacement mid-competition would be impractical.
C.3.	Must fit in a Skule™ Kup locker.	Packaged dimensions: L x W x H (cm).	$\leq 50 \times 50 \times 95$ cm	If the design exceeds the maximum dimensions of a Skule™ Kup locker, the client would not be able to transport it or store it, rendering the design inaccessible to its primary user.
C.4.	Must comply with the Ontario Electrical Safety Code (OESC). [19]	Number of exposed components.	0	From OESC Section 2-202: Guarding of bare or live parts.
		Number of dissimilar metals used in direct electrical connections.	0	From OESC Section 2-116: Corrosion protection for materials used in wiring.
		Deviation from rated voltage (%).	Within $\pm 5\%$	From OESC Section 8-102: Voltage drop.
		Operating current as percentage of any conductor, switch, or	$\leq 80\%$	From OESC Section 8-104: Maximum circuit loading.

		protection device's rated current (%).		
--	--	---	--	--

6.0. Conclusion

This report established the scope, context, crucial parameters, and measurable criteria for developing an effective connected buzzer system for Skule™ Kup. The report also explains where the design will be implemented, who will be using it and be impacted by it, as well as its limitations and purpose. The information in the report forms the knowledge foundation of the project and will be directly implemented in the Conceptual Design Specifications.

7.0 Reference List

- [1] “Skule™ - Directorships,” *Skule.ca*, 2024.
<https://skule.ca/get-involved/directorships.html> (accessed Feb. 14, 2026)

- [2] “Myhal Centre for Engineering Innovation & Entrepreneurship,” *Faculty of Applied Science & Engineering*,
<https://www.engineering.utoronto.ca/myhal-centre-for-engineering-innovation-entrepreneurship/> (accessed Feb. 14, 2026).
- [3] “*Reimagine the pit: Where possibility meets legacy*,” Engineering Alumni & Friends,
<https://alumni.engineering.utoronto.ca/thepit/> (accessed Feb. 14, 2026).
- [4] F. Pinto, “*Reimagining the pit, honouring its legacy*,” U of T Engineering News,
<https://news.engineering.utoronto.ca/reimagining-the-pit-honouring-its-legacy/#:~:text=In%20recognition%20of%20that%20value,to%20about%20the%20new%20space?>
(accessed Feb. 14, 2026).
- [5] “*Be a catalyst for change—here at U of T and around the world.*,” University of Toronto,
<https://engage.utoronto.ca/site/SPageServer?pagename=donate#/fund/2385> (accessed Feb. 14, 2026).
- [6] “*Archives*,” Campus Wireless,
<https://wireless.utoronto.ca/ufaq-category/connect-to-wi-fi/#:~:text=The%20eduroam%20network%20at%20U%20of%20T,visiting%20scholars%20from%20other%20participating%20eduroam%20institutions> (accessed Feb. 14, 2026).
- [7] “*5G/5G+, 4G LTE and HSPA+ network coverage map*,” Rogers,
<https://www.rogers.com/mobility/network-coverage-map> (accessed Feb. 14, 2026).
- [8] “*5G and 4G LTE, HSPA+ & LPWA network coverage map*” TELUS,
<https://www.telus.com/en/mobility/network/coverage-map> (accessed Feb. 14, 2026).
- [9] “*Space at the University of Toronto, procedures for the temporary use of*,” Office of the Vice-Provost, Students,
<https://www.viceprovoststudents.utoronto.ca/policy-guideline/space-at-the-university-of-toronto-procedures-for-the-temporary-use-of/> (accessed Feb. 14, 2026).
- [10] Petra.com, “*Digital Energy® Handheld Buzzer Game System Console with 8 Joystick Buzzers*.”
<https://www.petra.com/digital-energy-dee1-1143-handheld-buzzer-game-system-console-with-8-joystick-buzzers.html> (accessed Feb. 14, 2026).
- [11] “*Octobuzzy - enjoy buzzers*,” Enjoy Buzzers | Products | Enjoy Tech - Interactive

- specialists - Clickers, Wireless, Audience, Student, Trivia and Pub Quiz Response Systems, AV Concentrate Management System, Visualizers,
<https://www.enjoyvote.com/en-us/products.php?view=81> (accessed Feb. 14, 2026).
- [12] “Carry on baggage,” Air Canada.
<https://www.aircanada.com/ca/en/aco/home/plan/baggage/carry-on.html#/> (accessed Feb. 14, 2026).
- [13] “IEC 60068-2-31 rough handling shock testing,” Applus+ Keystone,
<https://keystonecompliance.com/iec-60068-2-31/> (accessed Feb. 14, 2026).
- [14] *1 B3F tactile switch through-hole-mounting switches in a wide range of models*., Omron.
https://omronfs.omron.com/en_US/ecb/products/pdf/en-b3f.pdf (accessed Feb. 14, 2026).
- [15] *Online materials information resource - matweb*, <https://www.matweb.com/> (accessed Feb. 14, 2026).
- [16] L. I. Labrecque and G. R. Milne, “Exciting red and competent blue: The importance of color in marketing - journal of the Academy of Marketing Science,” SpringerLink,
<https://link.springer.com/article/10.1007/s11747-010-0245-y#citeas> (accessed Feb. 14, 2026).
- [17] J. Plumb, “Engineering design principles & methodology - cambridge DT,” Cambridge Design Technology,
<https://www.cambridge-dt.com/a-deep-dive-into-engineering-design-principles-and-methodology/> (accessed Feb. 14, 2026).
- [18] *Buzzer systems*. [Online] <https://buzzersystems.com/> (accessed Feb. 14, 2026)
- [19] *Ontario Electrical Safety Code (OESC), 29th Edition - 2024*. CSA Group.
<https://www.csagroup.org/store/product/CSA%20C22.1%20ON%3A24/> (accessed Feb. 14, 2026)

8.0 Appendices

8.1 Appendix A: Engineering Notebook Notes

A.1 Screenshot of Omo's client meeting #1 notes

Client Meeting #1 notes

- 2 hours of battery runtime
- no electrical/fire hazard
- able to engage/disengage
- director and project director
- no jamming/beeping for lot seconds
- locker in the Mythical Arena
- usb-c connection
- repairability
- LED indicator for each buzzer
- plywood, metal, 3D printer
- IP66 resistance
- 48 hours



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

DATE _____

STUDENT NO. _____

PG

FULL NAME

ASSIGNMENT

COURSE

- week after reading week for 2nd meeting

A.2 Screenshot of Kamilia's client meeting #1 notes

Client meeting Notes (Cleaned up)

January 29 2026 - Galbraith Lobby @ 1pm

Client: Alexandre Klaus

Members Present: Henry, Omo, Shannon, Merrick, AS

- o Client cares about the buzzer being compact and portable of all things, one of the top priorities for him
 - ↳ specifically 30x30x30 cm is a good objective for him
- o Usability comes first for the design, then compactness and portability. (How well the buzzer works)
- o Does not have a preference on the power source for the buzzer.
 - ↳ battery, plug, etc. Not a top priority.
 - ↳ Service environment has a lot of plugs
- o The budget for the design is around 250\$
- o The client is indifferent to if the design makes any light or sound (is not a top priority, could be an objective)
 - ↳ liked the idea of the buzzer lighting up more than making sound
- o The design is specifically wanted for discipline fued. Encouraged us to focus our design on that game
- o The games typically run for 2 hours, so the buzzer should be able to run for more than two hours (if battery is used)
- o The main reason commercial buzzers aren't meeting his needs is because they aren't big enough or have enough buzzers
 - ↳ because discipline fued has up to 8 disciplines playing needs a big amount of buzzers
- o Would like handheld buzzers that shouldn't be heavier than a waterbottle (1.1kg is his estimate)
- o Client recommends wires and not bluetooth
 - ↳ Because of signal issues that comes with wireless devices. But we can still explore the option
- o The Client wants the buzzer to have a lock out system, where the participants won't be able to answer until the question is answered
- o The design should be able to withstand falling from a table height or a little lower (20 cm is an estimate)
- o Ideally the materials can be made from MYFab materials
 - ↳ Client suggests 3D printed parts as a goal
 - ↳ Plywood as well
 - ↳ Preferably no metal since that's heavy
- o Make sure the design has no fire hazards (shocking)
- o Interest holders: future Skule Kup director (when appointed), Alexandre Klaus
- o The design should fit in the locker where it's kept
- o Should be water resistant
 - ↳ IP66 water resistance (according to client)
- o Skule Kup discipline fued:
 - ↳ 8 teams, separated by discipline
 - ↳ normally there's about 10-40 people that attend the games
 - ↳ indoors at the pit of GB (under construction, this year it's held at Suds)

A.3 Screenshots of Omo's site visit notes.

February 14th, 2023

EngSec Arena Site Visit

- 17.2m garage to garage width
- 16.75m garage to stairs length
- Outlets: 4 on each garage wall (12 total)
- Tables: Shorter = 0.85m, taller = 1m (9 total)
- Each garage is 2.5m wide and dedicated to a specific design team
- Temperature: 21°C
- Signs: Club names, emergency exit, safety board, First Aid



UNIVERSITY OF TORONTO
FACULTY OF APPLIED SCIENCE & ENGINEERING

DATE

STUDENT NO.

PG.

5

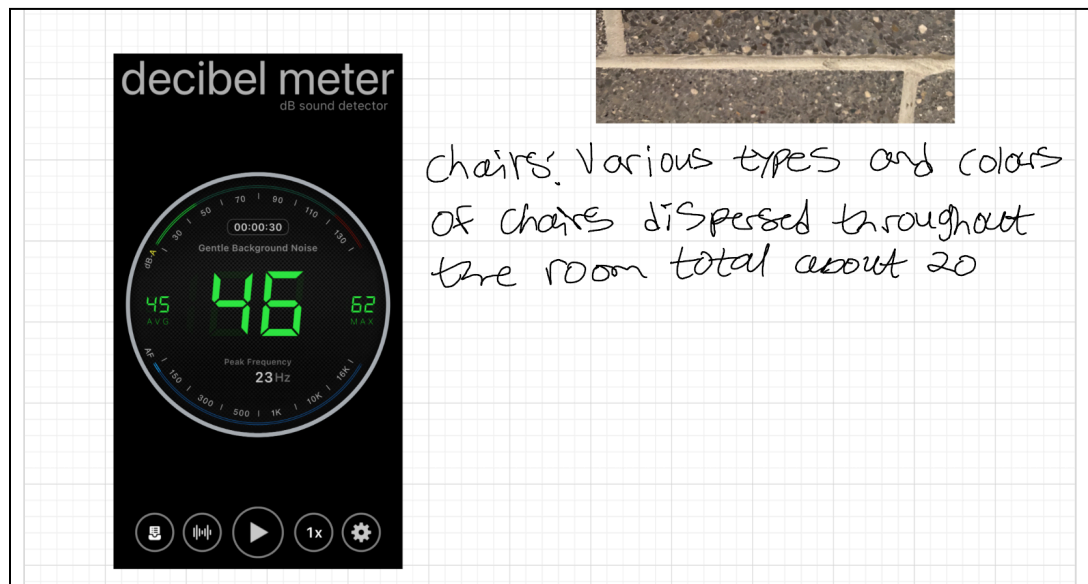
FULL NAME

ASSIGNMENT

COURSE

- 2 garbage units/fire extinguishers





March 2nd
3rd EngSOC site visit

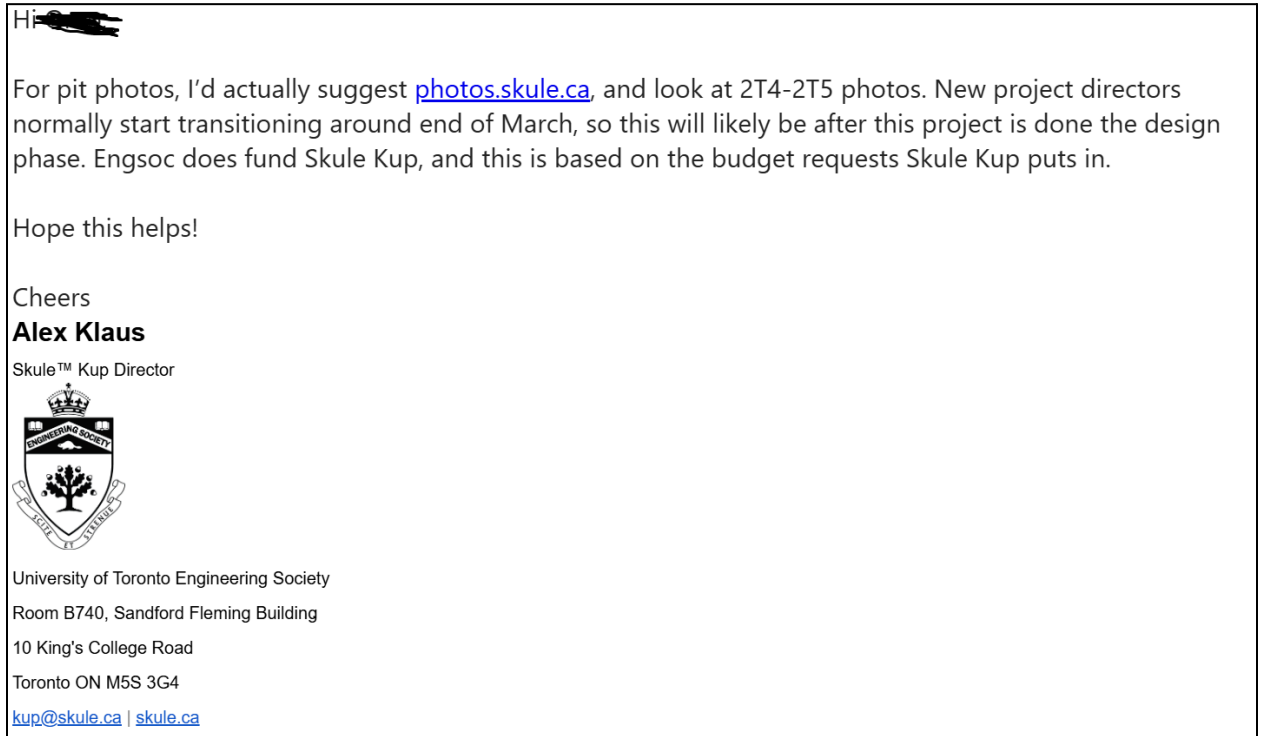
- 7/7 U of T eng students
• 4 girls, 3 guys

8.2 Appendix B: Client Emails

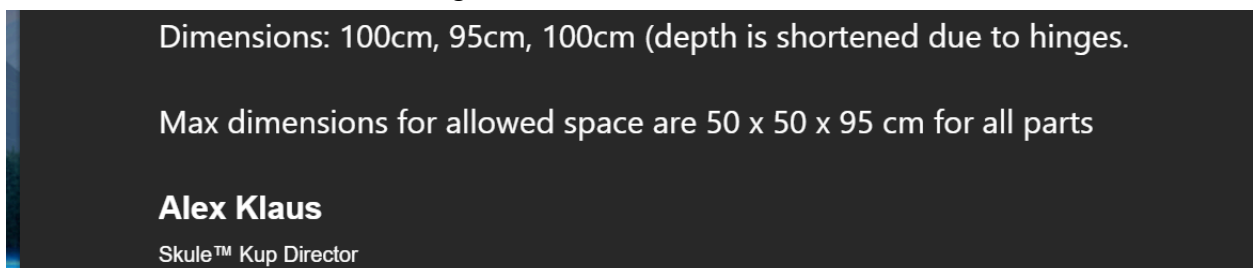
Appendix B contains screenshots of the second client email that was sent to the team and the email confirming the size of the locker the design will be stored in.

☐

B.1. Screenshot of second client email



B.2. Screenshot of email confirming the size of the locker.



8.3 Appendix C: Interest Holder Ranking Justification

Appendix C includes the justifications for ranking the interest holders for our project from most to least important.

C.1. The impact/influence matrix that was used to rank interest holders' importance.

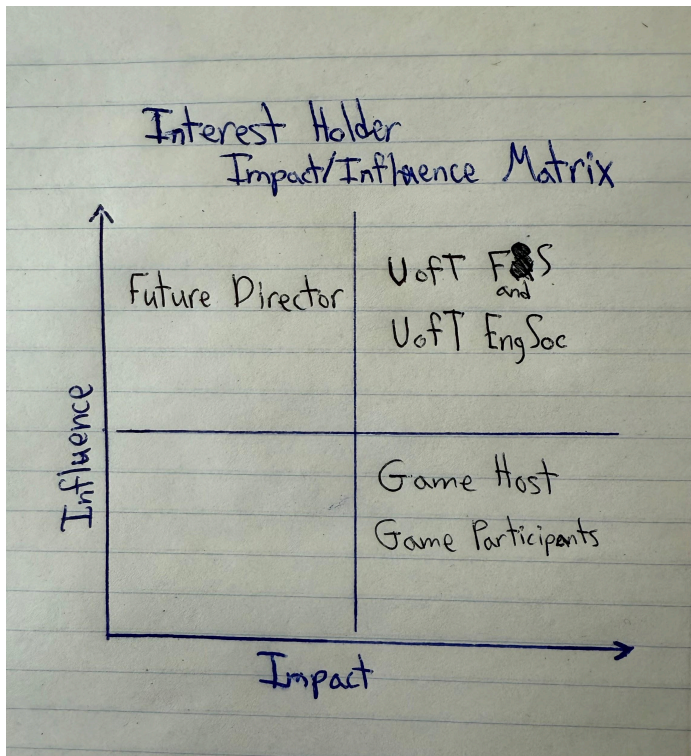


Table C.1. includes the justification for each interest holder placement in the impact/influence matrix.

Table C.1. Justification for the interest holders of the project

Interest Holder	Impact/Influence
University of Toronto Facilities & Services	Highest influence. They can approve or deny use of the system in campus spaces, which directly determines whether it can be deployed at all.
University of Toronto EngSoc	High financial and organizational influence through funding and oversight of Skule™ Kup. Their approval affects whether the system is supported and used.
Future Skule™ Kup Director	Moderate-high operational influence over future adoption and continued use of the system across years.
Discipline Feud Game Host	Moderate-high impact. The host interacts with the system during gameplay and is affected by its functionality, but does not control funding or approval.
Discipline Feud Participants	Moderate impact. Competition results depend on the system's response recognition and participants are affected by its performance during gameplay, but they have no control over funding and approval.

8.4 Appendix D: Expansion on the functions

Appendix D shows the justifications for both of our primary and secondary functions.

Table D.1. Contains an in-depth outlook into the functions

Primary Function	Justification
The design will enable the game host to identify information from the audience over a distance	The primary function transfers responses from a distributed set of players to a single host. This is the primary function because identifying the audience response attempt ensures fairness and avoids confusion in the Skule™ Jeopardy game, where response attempt orders are matched accordingly [17].
Secondary Functions	Justification
Project the signal of the response attempt from the audience onto the display screen	The projection of the response signal to alert the host to the information enables fast identification, regardless of the physical distance between them. This ensures that response outcomes are clearly communicated to the host, mitigating confusion and interpretation subjectivity.
Capture/store player input and information from the physical pressing of the buzzer by the players	Capturing and storing player inputs and information enables the primary functions by ensuring reliable data is being transmitted to the host. As such, this prompts accurate identification of who responds first.

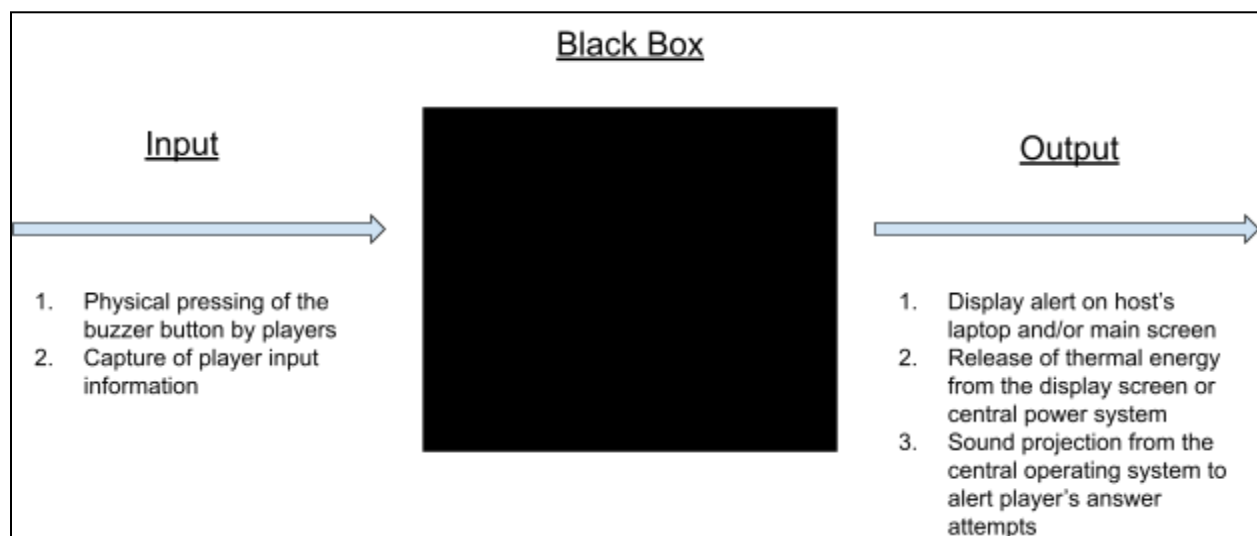


Figure D.2. Analyzing the functions through the black box method

8.5 Appendix E: Details on the objectives.

Appendix E contains the justifications for the objectives and the pairwise method used to justify the ranking of the objectives.

Table E.1. Justification table for the objectives of our design.

Objective	Justification
O.1.	Handheld consumer devices such as gaming controllers, buzzers, and microphones typically weigh 0.2-0.5 kg, preventing user fatigue during extended use. The metric comes from an average of multiple different devices that all weigh about 0.5 kg for each buzzer and 1.6kg per system.
O.2.	Standard airline carry-on luggage limits for personal items are approximately 13 x 6 x 17in (Transport Canada Air Travel Guidelines; major airlines such as Air Canada). Designing the full buzzer system to copy those dimensions ensures it can be easily stored in a locker, and carried by one person without logistical burden. Which is a need stated by the client (Appendix A). Inches were converted to centimeters, for consistency.
O.3.	- IEC 60068-2-31 tests portable electronics at 1 meter drop height. This goal is also valid since the height of a table in the service environment of our design is around 1 meter high. - Commercial tactile switches are commonly rated for at least 100 000 actuations before mechanical failure, as specified by manufacturers such as Omron and Panasonic. - ABS and Polycarbonate are widely used in electronics housings due to their high impact resistance. MatWeb's database on materials show that a high impact strength is from 60-80 kJ/m².
O.4.	The client specified that the design should be predominantly made out of MYFab materials. The 80% MYFab fabrication goal aligns with scholarly recommendations that undergraduate design projects maximize in-house fabrication to improve hands-on design learning.
O.5.	Comparable 4-8 player commercial buzzer systems retail between \$200-\$400, the target cost keeps the design feasible and economically competitive.
O.6.	Brand recognition research shows that dominant colour coverage above 70% improves visual association and identity recognition. Additionally, a high amount of said colours makes the design more catered towards Skule™ identity.

8.6 Appendix F: Constraints justification

Appendix F displays the justifications for the constraints of our design.

Table F.1. Justifications of the constraints of the design.

Constraint	Justification
C.1.	The client requires a system that can be used by a minimum of two users simultaneously, as Skule™ Kup competitions require multiple participants per session (Appendix G).
C.2.	The client requires a design suitable for Discipline Feud at Skule™ Kup, with typical sessions estimated to run for two hours (Appendix A). A runtime exceeding two hours is therefore necessary to prevent operational interruptions during gameplay. Failure to meet this constraint would make the system functionally incomplete, as recharging or replacement mid-competition would be impractical.
C.3.	The client, as Skule™ Kup director, will be storing the system in a Skule™ Kup locker between events. If the design exceeds the maximum dimensions, the client would not be able to transport it or store it, rendering the design inaccessible to its primary user (Appendix B).
C.4.	As a device handled by students, the design must comply with safety standards to ensure general user protection. The Ontario Electrical Safety Code [19] is the governing regulation for electrical equipment in Ontario. The C.4. sub-constraints were specifically drawn from the following OESC sections, to address the hazards most relevant to a small-scale electrical system: Section 2: General Rules ● 2-116 Corrosion protection for materials used in wiring: ○ Exposed terminals must be insulated or closed. ● 2-202 Guarding of bare or live parts: ○ Dissimilar metals shall not be used where there is a possibility of galvanic corrosion. Section 8: Circuit Loading and Demand Factors ● 8-102 Voltage drop: ○ Voltage supplied to each component must be within $\pm 5\%$ of rated voltage to ensure proper operation. ● 8-104 Maximum circuit loading: ○ The system's operating current shall not exceed 80% of the rated current of any conductor, switch, or protection device during continuous operation.

E.2 Screenshot from Kamilia's engineering notebook of the brainstorming of the objectives.

February 6-10 2026

Brainstorm of Objectives

- o Compact, portable, objective: $30 \times 30 \times 30 \text{ cm}$ → from client (do own research)
- o Powered by a battery, objective: }
- o Powered by being plugged into a wall: } not neces
- o portability: objective:
- o makes a light
- o makes a sound (likes this more)
- o lasts 2 hours, durable for at least a year X → constraint
 ↳ if it's battery
- o handheld buzzers: objective: weigh about a waterbottle (1Kg)
- o Should be durable (should be able to withstand falling off a table) objective: 20cm
- o Should be made from MYfab materials, objective: 100% (3D printed, no metal, plywood, plastic) ↑ from client (do research)
- o Cost effective: objective: 250\$
- o Should be water resistant, objective: IP67 or 68 (do research)
- o Should be the colours of Skule ↑ client suggestion

E.3. Screenshot of Kamilia's engineering notebook of the pairwise comparison box used to rank the objectives from highest to least importance.

	Compact	Portable	Time durable	Material durable	Represents Skule	MyFab materials	Cost effective	Total
Compact	X	0	1	1	1	1	1	5
Portable	1	X	1	1	1	1	1	6
Time durable	0	1	X	0	1	1	1	4
Material durable	0	0	1	X	1	1	1	3
Represents Skule	0	0	0	0	X	0	0	0
MyFab materials	0	0	0	0	1	X	1	2
Cost effective	0	0	0	0	1	0	X	1

8.6 Appendix G: Client Statement

Appendix G includes a screenshot of the client statement.

Engineering Strategies & Practice APS112 - Winter 2026 Project Description	
Project/Team:	13
CLIENT INFORMATION	
Client Name(s):	Alexandre Klaus
Professional Title, Occupation, or Identifier in relation to the project:	Skule Kup Director
Company:	
Address:	
Email:	kup@skule.ca
Telephone:	6479743544
Alternate Contact Name:	
Alternate Contact Email:	
Alternate Contact Telephone:	
Project Title:	Buzzer beaters: A portable buzzer system.
Need to develop a compact and portable buzzer system that indicates first responder in a jeopardy style. Should be able to have up to 8 buzzers connected, and as few as two. This would be used in certain Skule Kup events to allow for a better format to things like Discipline Feud, the typical Skule Kup closing event.	